Foundations of Data Engineering and Analytics

Department of Computer Science
University of Innsbruck

Wolfgang Gassler, Eva Zangerle

Data Analysis Project

Throughout the semester, you are going to work on a larger data analysis project. This will involve the full data analyis workflow—from data gathering, cleaning and preprocessing, to exploratory and confirmatory analysis, modeling and evaluation.

1 Overview

Note that this project is worth 30 points. Thus, we expect you to provide a comprehensive analysis, method development, evaluation, and documentation in the course of the project.

You may work on the project either individually or in groups of a maximum size of three people. Note that also for the project, the same rules for group work apply (see guidelines on OLAT).

Feel free to use any of the methods you learned throughout the lecture and tutorial, but also further methods that were not part of this course.

After your final submission, we will go through all your submission files and adjust the points reached if necessary.

2 Project Goal

For this project, we ask you to freely choose a research question to tackle. The only requirement is that the intended research question allows you to cover each of the tasks stated in Section 3. In the following, we list a few example research questions:

- How can the overnight stays in Innsbruck be predicted/forecasted?
- How can we predict the chart rank of songs based on high-level audio features?
- How can we predict the number of future purchases of electronic vehicles?
- What customer behaviors (clicks, time on site, purchases) predict whether a user will make a purchase in the next session?
- How can we predict student performance based on aspects such as parental education, background, or behavioral factors?

3 Project Contents

We expect you to tackle the full workflow of data analysis in the course of this project. Table 1 provides you with a high-level overview of the project tasks and the points awarded for each of the tasks. Furthermore, we provide you with a list of possible actions (subtasks) to take during the task.

Task	Subtasks	Points
Concept	Clear research goal and description how you will achieve the set goal	4
Data processing	Dataset description, data gathering, cleaning, preprocessing, bias mitigation, sampling strategies	4
Exploratory data analysis	Graphical and descriptive uni-, bi-, and multivariate analysis	6
Modeling and evaluation	Modeling methods, evaluation setup, metrics, baselines, tuning	10
Hypothesis testing	Null/alternative hypothesis, test statistics, conclusion/interpretation	2
Interpretation of results	Appropriate evaluation methods, reflection of used methods	4

Table 1: Overview of Project Contents

3.1 Concept

To make the goals of your project clear for us and to be able to provide you with feedback on your whole project as early as possible, we ask you to prepare a concept. We expect this concept to be roughly two pages and be prepared to present it in one of the first presentation sessions (see Table 3). The concept should contain the following information:

- Which research question(s) do you intend to investigate and answer?
- Why is this question interesting and challenging?
- How does the project relate to existing works?
- Which data is required?
- How can you retrieve such data in sufficient quantity and quality?
- Which methods and resources will you use for the analysis, prediction, and evaluation?
- What could be potential risks and limitations of your project?

3.2 Exploratory Data Analysis

Focus on your defined research goal and the upcoming modelling steps when exploring your data set. Please provide an explanation why and how the data analysis contributes to better results and if the newly gained insights require further data processing. Also, please report how the results of the exploratory data analysis inform the feature engineering and modeling steps that are up next. Note that the results of the EDA might also require you to go back to the data processing step and fine-tune the dataset.

3.3 Modeling, Evaluation, and Hypothesis Testing

Note that for the modeling, evaluation, and tuning of your approach, we expect you to use and compare different methods, feature engineering approaches, and parameter settings to tune your approach. Provide reasoning on your decisions, particularly why you have chosen a particular model. You should be able to demonstrate that you did extensive experiments to validate your approach. Also make sure to include sensible baselines to contextualize the evaluation results.

3.4 Interpretation of Results

Please answer your research question(s) concisely and comprehensively. Provide enough explanation of why you achieved these results and which implications they have for future work.

4 Important Dates

The deadlines for the individual project parts are listed in Table 2. All parts have to be submitted via OLAT before 23:59 on the stated date. The intermediate deadlines are meant to ensure an even workload throughout the semester. Your final submission will be used for grading the project.

Date	Content to be Submitted via OLAT
27/10/2026	Concept (appr. 2 pages; either pdf or ipynb and html file) Dataset pre-processing (ipynb and html file) Datasheet for the dataset[1] (pdf)
24/11/2026	Exploratory data analysis (ipynb and html file)
19/01/2027	Final submission: • Concept (pdf) • Datasheet for the dataset (pdf) • team.txt (txt) • Overall approach and evaluation results including data set (csv) and Jupyter notebooks (ipynb and html files)

Table 2: Overview of Important Dates

Please note that your notebook(s) should provide a self-contained presentation of what you did, which sources you used, and which methods you used (also including potential explanations of approaches, references to underlying papers, reasoning about your decisions, etc.).

The presentations of the first two milestones are spread over several tutorial sessions (see Table 3), i.e., in each of these sessions some of the groups present. Which groups present in which session will be announced in the tutorial. Every member of the group should be able to individually present the group's milestones to the others. This presentation should comprise an overview of your approach, evaluation setup, results, and further observations that might be interesting to present.

Dates	Presentation Content
04/11, 11/11, 18/11/2026	Concept incl. research question and data set (appr. 5 minutes per group)
02/12, 09/12, 16/12/2026	Results of exploratory data analysis (appr. 5–10 minutes per group)
20/01/2027	Final project (appr. 10 minutes per group)

Table 3: Overview of Presentations

Questions?

Please contact eva.zangerle@uibk.ac.at (group 1) or wolfgang.gassler@uibk.ac.at (group 2) in case of any questions or if you need feedback on your approach.

Important: Submit your solution to OLAT and mark your solved exercises with the provided checkboxes. The deadline ends at 23:59 on the day before the discussion.

[1] A datasheet documents the dataset you use: its motivation, composition, collection process, preprocessing, intended uses, and limitations. See Gebru et al., *Datasheets for Datasets*, https://arxiv.org/abs/1803.09010, for the structure and guiding questions.